

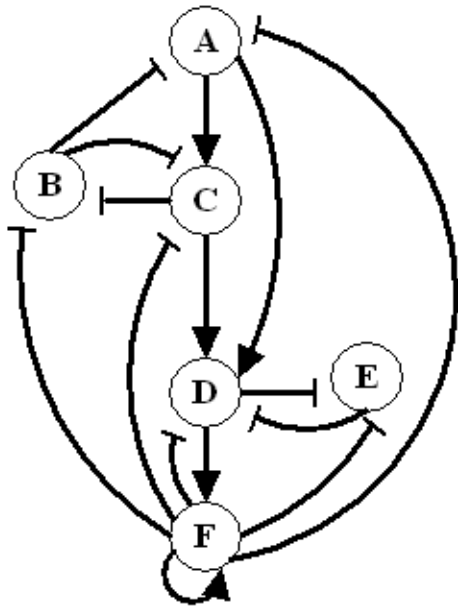
$$A^* = B$$

$$B^* = A \text{ and not } C$$

$$C^* = B$$

Ex. 7 Find the state transition graphs corresponding to

1. Synchronous update
2. General asynchronous update (update one node at a time)
3. Deterministic asynchronous update with $t_A=6$, $t_B=3$ and $t_C=2$



$A^* = \text{not } B \text{ and not } F$

$B^* = \text{not } C \text{ and not } F$

$C^* = A \text{ and not } B \text{ and not } F$

$D^* = (C \text{ or } (A \text{ and not } E)) \text{ and not } F$

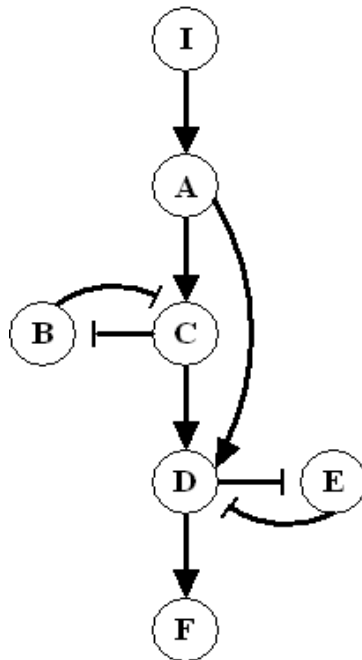
$E^* = \text{not } D \text{ and not } F$

$F^* = D \text{ or } F$

Ex. 8. Consider the 6-node network corresponding to T-LGL leukemia. Recall that this network has two fixed points and no complex attractors. Recall that according to Naldi et al. 2011 collapsing any node with no self-loops preserves the fixed points.

- (a) Propose ways to simplify this network further.
- (b) Determine the state transition graph of the simplified network using general asynchronous update.

Ex. 9. Consider this variant of the T-LGL network. I is an input signal to the network and F is the output node.



$$A^* = I$$

$$B^* = \text{not } C$$

$$C^* = A \text{ and not } B$$

$$D^* = C \text{ or } (A \text{ and not } E)$$

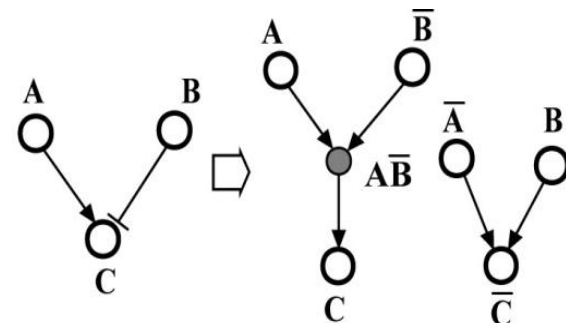
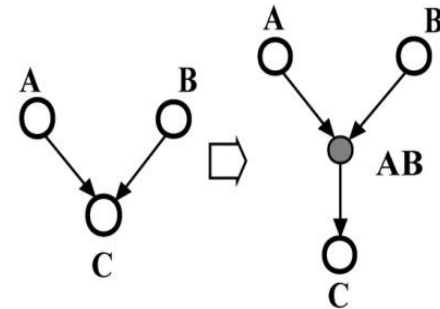
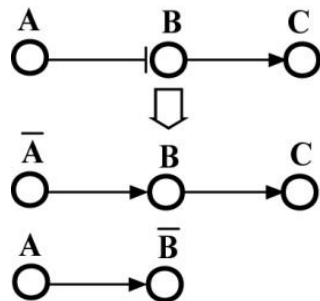
$$E^* = \text{not } D$$

$$F^* = D$$

- Expand the network by the addition of complementary and composite nodes.
- Determine the number of elementary signaling modes between I and F

Recall: Network augmentation

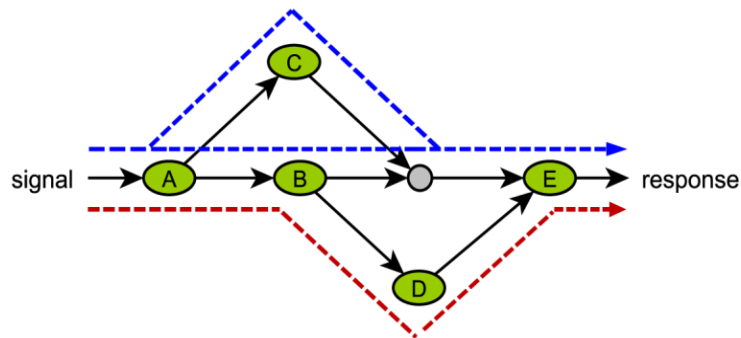
- Gather evidence of conditionality
- Add composite nodes for conditional relationships
- Add complementary nodes for nodes that inhibit/ are inhibited



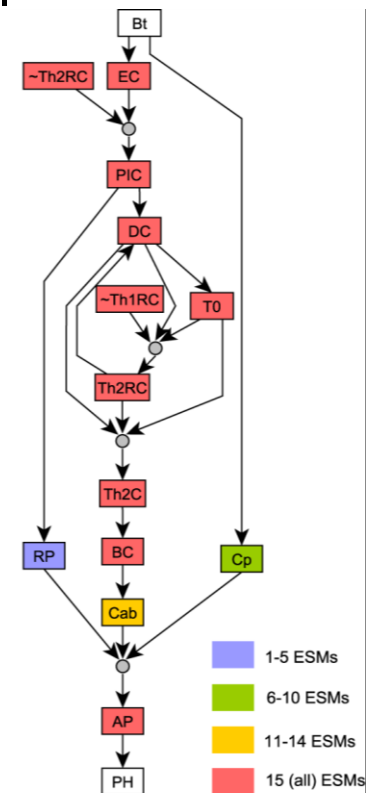
All edges that end in composite nodes are conditional, edges that end in original nodes are independent.

Recall: Elementary signaling modes

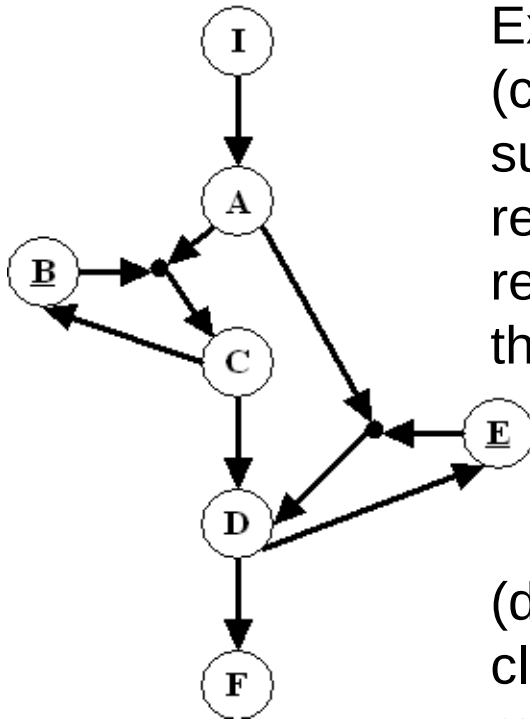
- ESM: a minimal set of nodes that can mediate signal transduction from input to the output.
- If an ESM contains a composite node, it must contain all of its regulators as well



- ESMs can overlap



Shortest ESM in immune response network



Ex. 9 continued

(c) Determine the essentiality of each node in this sub-network. Recall that the essentiality value is the relative change in the number of ESMs after removal of the node and after taking into account the cascading effects of the removal.

(d) Verify one node classified as essential and one classified as non-essential. Implement the node removal as a knockout or overexpression, i.e. removal of A is equivalent with $A=0$ and removal of B is equivalent with $B=1$. Assume $I=1$ and determine the fixed points of the Boolean model. If a fixed point with $F=1$ exists, the removed node is non-essential.

Recall: The effects of node loss cascade through the network

Removal of an original node simulates knockout.

Removal of a complementary node simulates overexpression.

Loss of a required activator leads to the loss of the target node.

Required activators:

any activator of a composite node,
sole (remaining) activator of
an original or complementary node.

